# SODIUM TECHNOLOGY IN JAPAN

S. KAWAHARA,* K. FURUKAWA† and K. SAKO†

## 1. Introduction

Energy consumption in Japan has risen rapidly in recent years. Resources such as coal, oil, and fissile materials are far from sufficient for future needs. For this reason, we take a great interest in atomic power generation, especially in efficient breeder power reactors. It is expected that the sodium cooled fast breeder reactors will be in use earlier than breeder reactors of other types. Consideration is now being given to the construction of an experimental or prototype reactor of this type in the near future as part of the fast breeder development programme. As a necessary step for this development, sodium technology has been studied in Japan since 1957.

Studies on sodium technology were started by Hitachi Ltd., and Japan Atomic Energy Research Institute (JAERI) participated in this activity a little later sending their staff members. Several sodium loops of 25–76 mm piping, with heat generations up to 200 kW, have been constructed by Hitachi Ltd. These have been operated jointly by the two organizations in various studies on corrosion, impurity control, heat transfer, and small scale sodium component tests. Reactor grade sodium, for these loops, has been supplied by Showa Denko Co. Ltd. One of the loops was transferred to the JAERI site in 1965.

To establish sodium technology for the development and construction of sodium cooled fast reactors, an out-of-pile sodium test loop of fairly large scale is planned at present, based on experience gained so far. The heating capacity of the main heater at the maximum temperature of the loop is about 2 MW, the loop therefore being called the 2 MW loop. The purpose and the characteristics of this loop will first be described, and then various earlier results obtained by other loops discussed.

* Hitachi Research Laboratory, Japan.
† Japan Atomic Energy Research Institute, Japan.

## 2. 2 MW Sodium Loop

### General Outline

The primary purpose of this loop is to examine various characteristics of components for an experimental fast reactor to be built in the near future. It is also expected to supply valuable engineering data for the cooling systems of future large scale fast breeder reactors. Domestic equipment and other facilities will be used for most of the loop system. The design features of the loop are as follows.

The maximum nominal temperature and pressure of the main loop are 650°C and 10 kg/cm$^2$, respectively. The diameter of the main pipe line is 150–200 mm. The a.c. 3 $\phi$ immersion type main heater is rated at 2 MW for 650°C sodium and 3 MW for reduced temperatures. The output of the main air cooler is 3 MW for 500°C sodium. The capacity of the main circulating pump of the free surface centrifugal type is 180 m$^3$/h for a head of 8 kg/cm$^2$ and 300 m$^3$/h for reduced heads; the maximum pump inlet sodium temperature is 450°C.

### Test Items

The components to be tested are as follows:

(i) *Endurance test of core fuel assemblies*

In order to freeze the design of the core fuel assemblies, dynamic flow and endurance tests in high temperature and high flow velocity environments are necessary. In the case of the endurance test, the temperatures used are 600–650°C and flow rates 1·5–2 times that in the actual core fuel assembly. In testing the core fuel assembly for our experimental fast reactor, flow rates of about 60 m$^3$/h and pressure differences of about 5 kg/cm$^2$ are necessary. In the case of the core fuel assembly for future commercial fast breeder reactors, a flow rate of 18–300 m$^3$/h and pressure differences of about 6 kg/cm$^2$ will be required. The rated output of the main circulating pump is designed to take these conditions into consideration. Since the maximum temperature of the main pump is 450°C, a regenerative heat exchanger must be used to minimize the heat input. In Fig. 1 the schematic diagram for endurance testing a large scale core fuel assembly is shown. In the case of small scale assemblies, such as in our experimental fast reactor, this loop can test three to five assemblies at a time with a regenerative heat exchanger and one assembly without the exchanger.

(ii) *Heat transfer experiments on full size core fuel assemblies*

In order to establish the design criteria of the core, it is necessary to clarify the heat transfer characteristics of the core fuel assemblies. The characteristics at the transient conditions (power change, flow rate change, etc.)

and the boiling behaviour are especially important. We are going to test these characteristics by using a pin bundle with electric heaters inside (core fuel

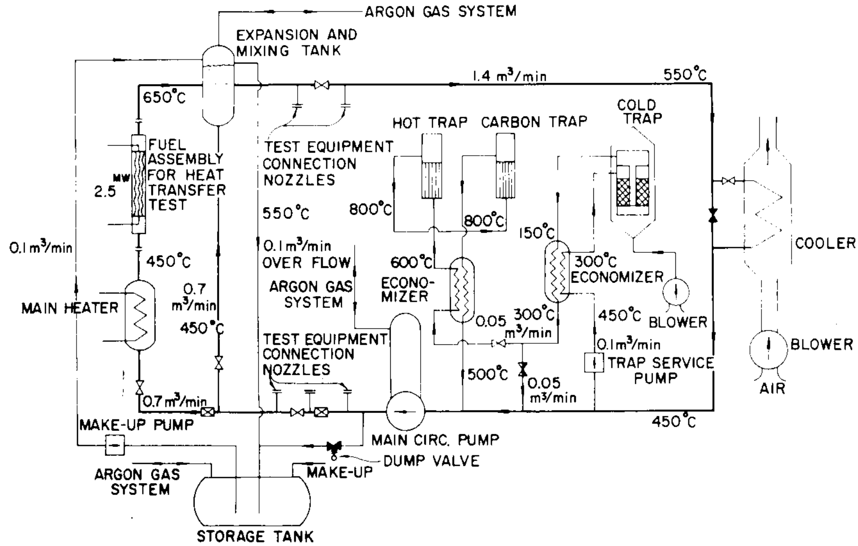


FIG. 1. Schematic flow diagram of the 2 MW sodium loop to test core fuel assembly heat transfer.

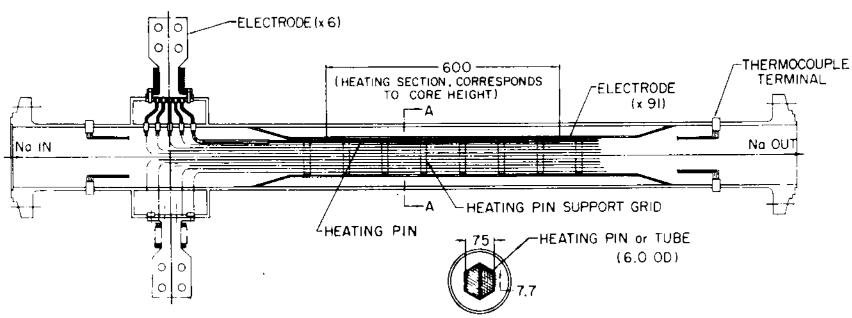


FIG. 2A. Electric heating pin bundle.

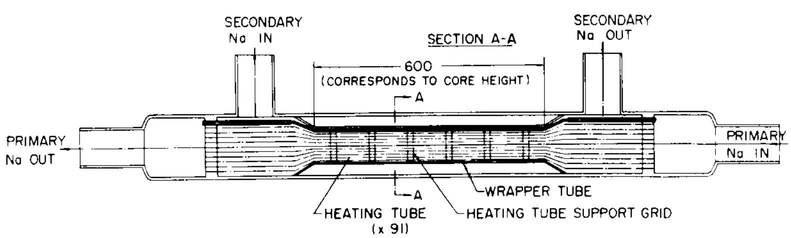


FIG. 2B. Heating tube bundle.

assembly shaped). Figure 2A shows one of our concepts of the electric heating pin bundle. The maximum nominal heat flux of the core fuel pin of our experimental fast reactor is about 250 W/cm$^2$; it is aimed to finally have the same heat flux. The heat transfer test is carried out by the connexion as shown in Fig. 1. The electromagnetic brakes are used for flow control and the main heater is used to control the assembly inlet temperature. In order to clarify the phenomena associated with the sodium boiling in the pin bundle at low temperature, the loop will be designed to withstand vacuum in the piping or other equipment. Because a heat flux of 250 W/cm$^2$ is not easy to obtain, the earlier phases of the experiment will be performed with fairly low heat fluxes. Figure 2B shows another concept of a bundle. The heating tube bundle is made from many small tubes, through which hot sodium flows. Because there are many difficult problems in measuring the temperature of the heat transfer surface in sodium, etc., there is much room for improvement.

(iii) *Tests on sodium components*

The loop can be used for the following sodium components test by modifications:

(a) *Tests for intermediate heat exchangers*

With the arrangement of the loop and heat exchanger, as shown in Fig. 3, the loop can be used for testing heat exchangers up to 10 MW, using the main heater. The transient conditions of the heat exchanger can also be reproduced by changing the temperature of the heater and of the cooler and controlling the flow rates by electromagnetic brakes. The time delay between the primary and secondary sodium circuits can be simulated by the surge tank, installed at the outlet of the primary circuit of the heat exchanger.

(b) *Tests for steam generators*

The testing of steam generators is also taken into consideration in the design of the loop. The 2 MW of a steam generator is too small, compared with that of the power reactors, but since the temperature is high enough, it is expected to be able to clarify the basic problems in high temperature and high pressure steam generators. It can test up to about 15 MW by adding another heater. The flow sheet is shown in Fig. 4.

(c) *Tests for large scale pumps*

Tests of various kinds of sodium pumps are usually carried out by using other test loops, but in the case of large scale pumps, heat generation in the pump and purification of sodium become important problems. In this case, the 2 MW loop can be used as a cooling and purification system by connecting with a pump test loop.

(d) *Testing the in-pile test loop*

It is highly desirable to have at least one in-pile test loop in our experimental fast reactor. This loop will have independent cooling and impurity treatment systems. In testing this loop, a heater output of about 2 MW at a temperature of about 600°C and a heat removal capacity of about 2 MW will be necessary.

(e) *Tests on several kinds of structural components*

Using an arrangement in which the by-pass line is connected to the inlet and outlet nozzles of the main heater and the test chamber is installed at the outlet of the heater, the loop will be able to determine the behaviour of structural components under stress.

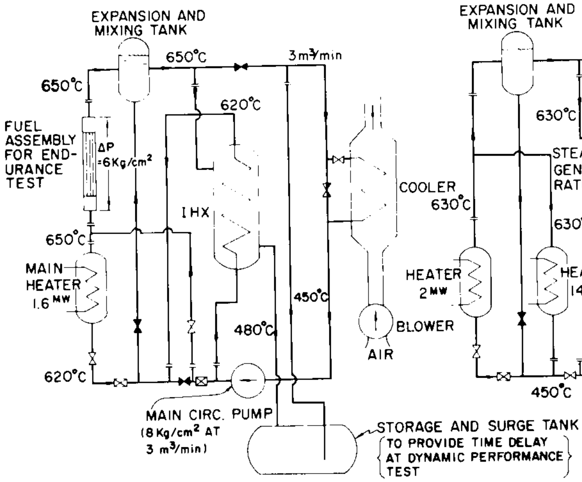


Fig. 3. Schematic diagram, showing intermediate heat exchanger and core fuel assembly endurance tests being performed.

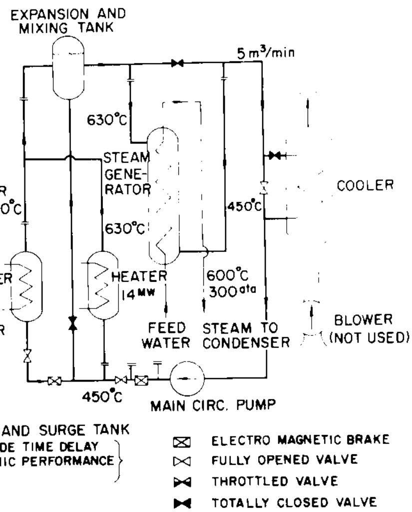


Fig. 4. Flow diagram for testing steam generator performance.

## 3. Engineering Background

### Small Scale Sodium Test Loops

We have had several small scale loops, which are as follows:

(i) HSL (Hitachi Sodium Loop)-1 was the first loop to be constructed as a general purpose out-of-pile sodium test circuit (200 kW). The primary purpose of this loop was to gain experience on the operation and maintenance of sodium loops. In addition, preliminary experiments on mass transfer phenomena were made with this loop.

(ii) HSL-2 (40 kW) was used for experiments on heat transfer, cold trapping, and impurity detection, as well as in tests of various instruments and small components.

(iii) HSL-3 (90 kW) was constructed as a mother loop, which supplies purity controlled sodium ($<5$ ppm oxygen) to several static corrosion test facilities and the small scale daughter loops. The purification of the sodium in the mother loop was done only by cold trapping.

(iv) JSL (JAERI Sodium Loop)-1 was a dynamic corrosion test loop, used for testing up to 600°C and 10 m/sec. A static corrosion test section installed in the expansion tank of the loop gave a comparison in the same sodium environment.

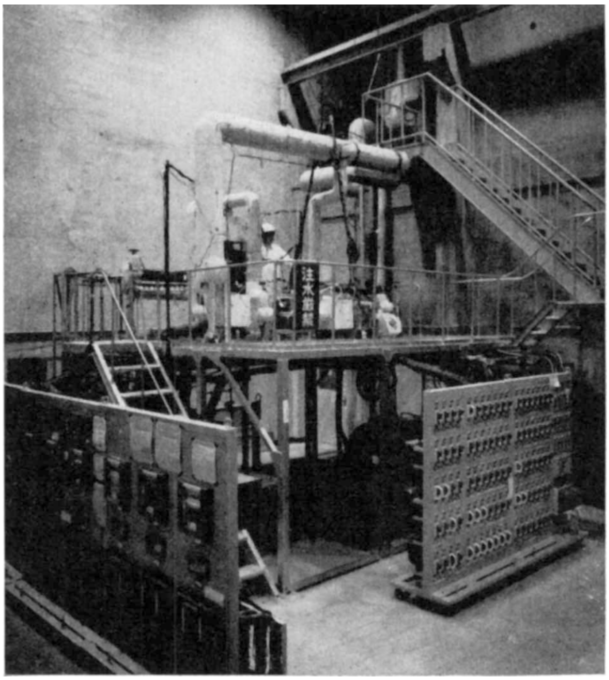

FIG. 5. JAERI Sodium Loop No. 3 (JSL-3).

(v) JSL-2 was a modification of JSL-1. The major modifications were the replacement of the cold trap by a larger one and the addition of a hot trap. This loop was used to examine the purification mechanism in the traps. Some results obtained from these experiments will be described later.

(vi) JSL-3 has recently been constructed as a mother-and-daughter loop system in the JAERI site. The mother loop supplies high purity sodium to the daughter loops in which various experiments can be made such as mass transfer, hydrogen separation and detection tests, flow calibration, etc. This system is shown in Fig. 5.

**Corrosion Tests of Structural Materials**

In order to determine the most economical sodium cooled reactor, it is important to establish the limits of usefulness of conventional materials. It is well known that austenitic stainless steel has a good compatibility with sodium and is a suitable material for fast breeder reactors.

Using JSL-1, the preliminary corrosion tests[1] were performed on the following materials: the type 304, 316, 347 stainless steels, $2\frac{1}{4}$Cr–1Mo steels, Nb, Ta, Mo, etc. The main purpose of these experiments was the preliminary development of sodium technology. The purity of the sodium could not be controlled sufficiently, and the longest duration time of the test runs was only 720 hours. In consequence, a large fluctuation in the data occurred, mainly due to the initial unsettled loop conditions. However, as a preliminary result, it was concluded there were no appreciable differences in the corrosion rate of the type 304, 316, 347 stainless steels, the average corrosion rate being about 0·14 mm/year in the following conditions: a maximum temperature of 600°C, a flow velocity of 10 m/sec, an oxygen content in sodium of about 30–20 ppm, a temperature difference of 400°C, and durations of tests of $400 \sim 720$ hours. The surface roughness after corrosion tests was about 1 $\mu$, the surface of 316 SS being somewhat smoother than the others.

Improved mass transfer experiments are now in progress, with the daughter test loops of JSL-3, under more closely controlled conditions.

Since the piping systems for high temperature sodium are constructed by welding, the corrosion behaviour of welded sections is also important. We welded thin and thick plates of austenitic stainless steel tubes, by several different methods, then corrosion tests were performed on the welded sections and heat affected zones. From these tests we found that different welding methods showed little difference from one another in corrosion behaviour of the welded sections, and that the corrosion behaviour in these regions differed little from that in the base metal. Of course, defects in welds bring about undesirable results.

Stress is another important factor in corrosion. It is probable that stress promotes corrosion and similarly the strength of a structure is affected by corrosion. Attention must be given to stress corrosion of cladding tubes of fuel elements and of tubes of a heat exchanger and a steam generator. In this connexion, creep tests in hot sodium were performed. The test pieces were 2 mm thick plates of type 304 stainless steel. The tank containing the samples was made of the same material and connected to the HSL-3. The tests were carried out for up to 1000 hours and the results on type 304 stainless steel are shown in Fig. 6. A chemical analysis and an observation by X-ray microanalyser of the exposed samples were performed. As indicated in Fig. 7 the effect of stress on the selective dissolution of Cr and carburization were measured. The latter was very small compared with the former.

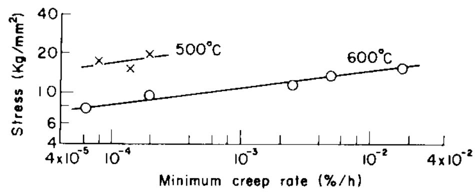


FIG. 6. Minimum creep rate of type 304 SS in hot sodium (plugging temperature 160°C).

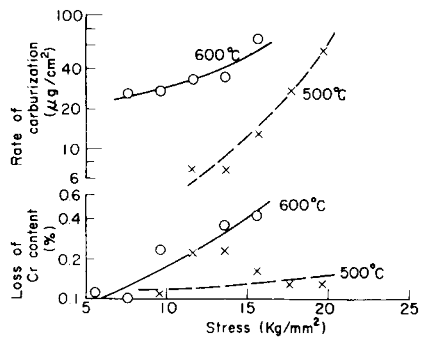


FIG. 7. Changes of Cr and carbon contents of type 304 SS stressed in hot sodium.

## Heat Transfer

The heat transfer in the regions of low flow velocities, and the boiling characteristics at the transient conditions, are considered to be very important in connexion with reactor safety. When coolant flow is lost or decreased these phenomena become critical.

We performed the heat transfer experiments in the low Péclet number range of from 5 to 250. Most of the measured Nusselt numbers started to fall at Pe $<200$ and in the region below 100 they were lower than the Lubarsky and Kaufmann's experimental formula of $Nu = 0{\cdot}625\ Pe^{0{\cdot}4}$ and were smaller than the theoretical value of $Nu = 4{\cdot}36$. The values measured lay between the extension line of the former and the curve described by Johnson *et al.*[2]

## Purification Methods

A cold trap of the type shown in Fig. 8 was investigated[3] in JSL-2 and JSL-3 and showed good performance. The effective inner diameter and volume of the trap cylinder were about 200 mm and 30 litres, respectively. The main features of its design are as follows:

(i) to prevent plugging in the dirty inlet path, a Hallam type helical inner-economizer, keeping a wide inlet path, is used;

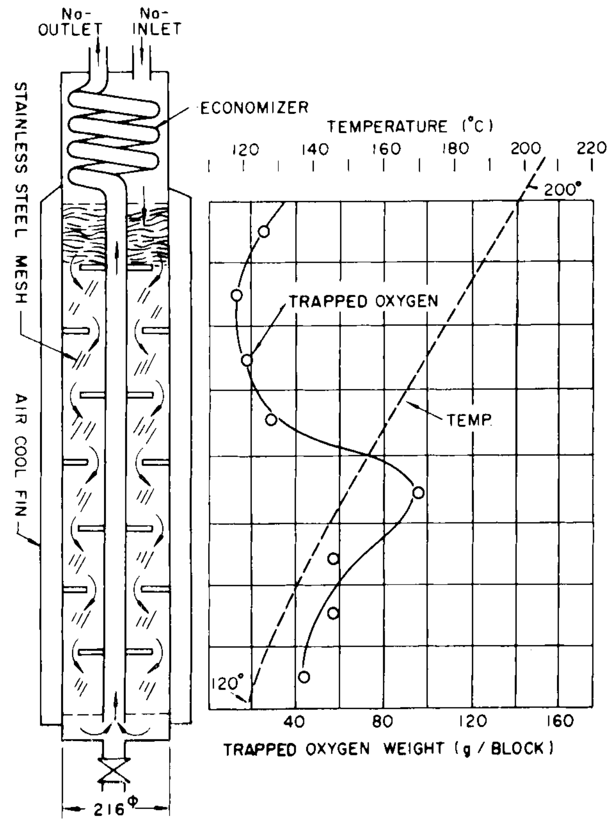


FIG. 8. Cold trap and the internal distributions of temperature and trapped oxygen.[3]

(ii) waved stainless steel wire nets of 30–40 meshes are packed coarsely at intervals of 3–4 mm. These wire nets have two or three randomly positioned holes. The whole assembly of wire nets consists of eight sections, supported by spacers, and they are separated by buffers from each other; the resultant paths of sodium are rather complex and relatively large. The packing density is $0{\cdot}18 \sim 0{\cdot}25$ g/cm$^3$;

(iii) the trap is cooled by means of forced-air cooled fins. The temperature distribution inside the trap is controlled by regulating this cooler and the flexible sheathed electric heaters wound on the fins.

After the experiment the cylinder was connected to a special glove box, into which the net assembly was pulled and examined by direct observation. Inserted representative specimens of wire nets were chemically analysed and their weight gain measured.

Sodium purification experiments, performed at the impurity levels of 10–50 ppm and a flow rate of about 4 l/min, showed that the amount of impurity, trapped before the trap plugged, was 340 g of oxygen, or 1·3 kg

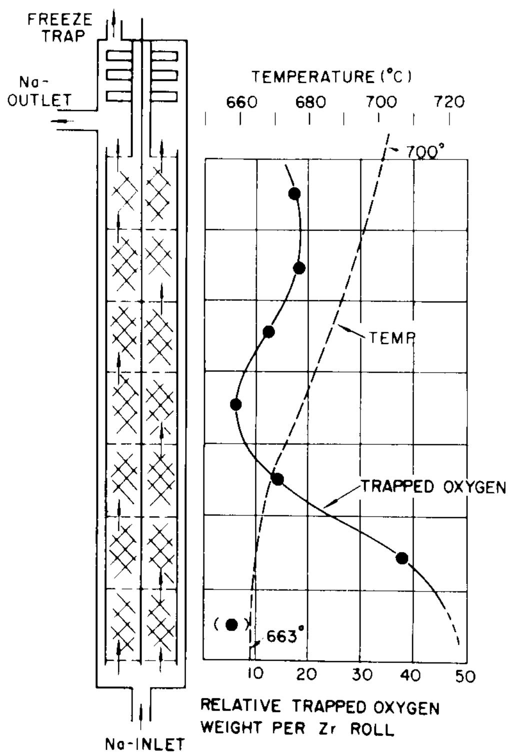


FIG. 9. Hot trap and the internal distributions of temperature and trapped oxygen.[4]

in the chemical form of $Na_2O$. The spatial distributions of the inner temperature used and of the trapped impurity are shown in Fig. 8. In other experiments at flow rates lower than 10 l/min it was shown that this cold trap had an impurity trapping efficiency of nearly 100%; this meant that the plugging indicator temperature for the outlet sodium was actually the same as the minimum cold trap temperature in the range of 250–130°C.

If an external heat economizer is added to the inner economizer and the cold trap temperature is controlled at the saturation level of the incoming sodium, better distribution will increase the trap capacity by a factor of 2 or 3.

A larger scale cold trap is now being designed, on the basis of the above results.

The test of a Zr hot trap was also performed in JSL-2.[4] A combination of a corrugated Zr-foil of 0·1 mm in thickness and flat foil of 0·05 mm in thickness was rolled, and eight such rolls were packed into a cylinder of effective volume of 3·85 litres (see Fig. 9). This cylinder had an electric furnace and an external heat economizer with a heater. Keeping the temperature distribution as shown in Fig. 9 in the trap, the performance was tested under the following conditions. The plugging temperature of the incoming sodium was 160–180°C, the flow rate 4 l/min, and the test duration 300 hours.

After completion of the above test, the hot trap was dismantled in a special glove box. The results of these experiments showed:

(i) the plugging temperature of the outlet sodium was 130–150°C;
(ii) the total amount of oxygen trapped was about 60 g for 3·23 kg of Zr foil;
(iii) part of the Zr foil became very brittle;
(iv) the apparent average oxygen-trapping rate was about 0·2 g oxygen/h or 15 mg/h $m^2$ of Zr surface; and
(v) the allowable maximum thickness of an oxide layer was concluded to be about 5 $\mu$.

The points which have to be modified to improve the performance are as follows:

(i) As shown in Fig. 9, the oxidation rate was too slow in the middle of the tray, where the temperature should be increased to enhance the oxidation.
(ii) To increase the effective area, the contact area between the corrugated and flat Zr foils should be decreased by modifying the corrugation.
(iii) The uniform sodium flow should be obtained by means of buffers and spaces between each roll. If the above improvements are made, the weight of oxygen trapped will be increased up to 70 g.

After purifying with such traps, the sodium in the loops was still very dirty because particulate suspensions in the order of microns or less in diameter still exist. This was shown by several results in the chemical analysis of carbon in sodium. Therefore, other purification methods are under investigation at present, including filtration, vacuum distillation, and diaphragm diffusion.

**Others**

Because of limited space, the description of the components used in these small scale loops and in the 2 MW loop, and of their experimental facilities, have been omitted. The details of the reactor grade sodium used is shown in Table 1.

TABLE 1. STANDARD FOR REACTOR GRADE SODIUM
(Showa Denko Co. Ltd.)

| Impurity | O | Ca | Si | Al | Fe | Mg | Cl | H | B | K |
|---|---|---|---|---|---|---|---|---|---|---|
| ppm | 10 ≯ | 5 ≯ | 5 ≯ | 5 ≯ | 10 ≯ | 15 ≯ | 15 ≯ | 5 ≯ | 0·01 ≯ | 2000 ≯ |

## 4. Conclusion

The experience gained in sodium technology is not yet sufficient. It is considered, however, that the results of tests, mainly from the 2 MW loop, will make it possible to plan for an experimental reactor and/or prototype reactor, and will give useful data for the development of equipment in their cooling systems, contributing to the overall development of fast breeder reactors.

The authors wish to thank Dr. M. Nozawa and Mr. T. Tamura of JAERI for their helpful advice.

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## Supplement

K. FURUKAWA

By the end of 1965, two 120-hour performance tests of the main part (mother loop) of JAERI's new test loop (JSL-3), which was completed in November 1965, were carried out.

During these operations it was possible to make a reliable measurement of the plugging temperature down to 110°C. It was confirmed that for a stainless steel loop of this size with a new surface, cleanup by sodium requires 3 days of operation at 400°C and 1 day at 500°C.

Two daughter loops, shown in Figs. S1 and S2, were added in 1966 and continuously operated from February 28 to March 19.

One of the two daughter loops is a mass transfer test loop which has loop piping of about 22 m of 15 mm i.d. × 2 mm thick, type 304 stainless steel tube, and has the total loop volume of about 18 litres. The loop contains the following components: an a.c. Faraday type pump, an electromagnetic flowmeter, immersion electric heaters (10+5+5 kW), an aluminium high finned stainless steel tube air-cooler.

The test specimens, 7·5 × 15 × 2 mm or 10 × 20 × 2mm, of type 304 and 316 stainless steel and $2\frac{1}{4}$Cr–1Mo alloy steel, were inserted into the six positions indicated as $H_1$, $H_2$, $H_3$, $C_1$, $C_2$, and $C_3$ in Fig. S1. The cross-sectional area at position $H_3$ is 10 mm and high velocities can be obtained.

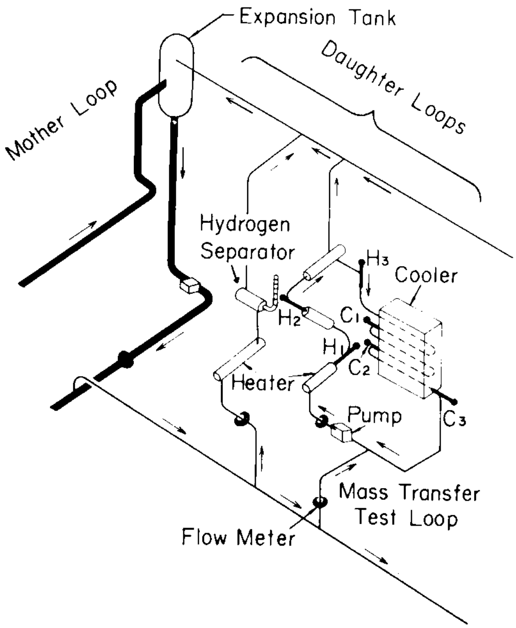


FIG. S1. Schematic figure of two daughter test loops (hydrogen separator and mass transfer test loops).

Initially, a preliminary cleanup of about 1 day was carried out by introducing 400°C sodium with a plugging temperature of 115°C from the mother loop which contained 420 kg sodium. Then, using the heater, the cooler, and the electromagnetic pump, 386 hours of test were carried out maintaining the temperature and flow conditions indicated in Table S1. The point of highest temperature, 600 ± 10°C, was near $H_3$ and the lowest temperature, 350 ± 5°C, was near the electromagnetic pump. The flow at the flowmeter was 4·4 ± 0·1 l/min and of this, 0·3 ± 0·05 l/min was continuously returned to the mother loop to maintain the plugging temperature at 116 ± 5°C.

The analyses are still being carried out. In Table S1, the weight changes for the average of four to five specimens are shown. Although the difference due to the steel types is not pronounced, it appears that the type 316 stainless

TABLE S1. MASS TRANSFER TEST CONDITIONS AND AVERAGE WEIGHT CHANGES OF TEST SPECIMENS (386 hours)

| | Na temperature (°C) | Na velocity (m/sec) | Weight change (mg/cm²) | | |
|---|---|---|---|---|---|
| | | | 304 SS | 316 SS | $2\frac{1}{4}$Cr–1Mo |
| $H_1$ | 495±15 | 0·56±0·02 | −0·003 | −0·003 | — |
| $H_2$ | 550±10 | 0·56±0·02 | −0·016 | −0·009 | — |
| $H_3$ | 600±10 | 6·8±0·2 | −0·14 | −0·11 | — |
| $C_1$ | 520±10 | 0·5±0·02 | +0·004 | +0·003 | — |
| $C_2$ | 460±8 | 0·5±0·02 | +0·06 | +0·014 | +0·022 |
| $C_3$ | 400±5 | 0·5±0·02 | −0·088 | −0·055 | −0·044 |

TABLE S2. EROSION RATE OF TYPE 316 STAINLESS STEEL AT MAXIMUM TEMPERATURE AND VELOCITY REGION (386 hours)

| Piping material | Max. temp. | Min. temp. | Flow velocity (m/sec) | Plugging temp. | Weight change (mg/cm²) | Reference |
|---|---|---|---|---|---|---|
| 304 | 600±10°C | 350±5°C | 6·6 | 116±5°C | −0·11 | This work |
| 316 | 600 | 316 | 6·4 | 147 | −0·07 | (S1) |
| 316 | 650 | 510 | 6·0 | 149 | −0·15 | (S1) |
| 316 | 704 | 427 | 5·5 | 232 | −7·25 | (S2) |

steel is superior to type 304. A possible material for the evaporator section of a steam generator $2\frac{1}{4}$Cr–1Mo steel was placed in the cold side and showed good results.

It is of interest to compare these results with the data of GEAP.[S1,S2] For a type 316 stainless steel loop, the 386-hour interpolated values for the maximum temperature and flow position are shown in Table S2.

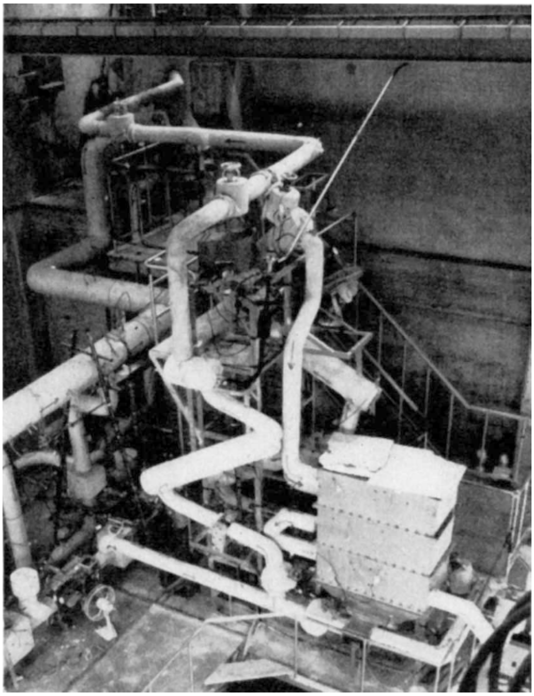

FIG. S2. General view of two daughter test loops (hydrogen separator and mass transfer test loops).

During the loop experiment, the carbon content of the sodium which was initially $73 \pm 8$ ppm decreased to about $41 \pm 3$ ppm. No carburization was observed microscopically in the test specimens. It is possible that the major part of carbon was used up in the carburization of the 700°C hydrogen separation test loop described in the next section. Further tests will be needed to substantiate this.

The other daughter loop is for the detection and the separation of hydrogen. As can be seen from Figs. S1 and S2, the loop consists of an immersion heater, a flowmeter, and a hydrogen separator. The hydrogen separator is a bellows, 34 mm o.d. × 190 mm, inserted in a cylindrical shell, 89·1 mm i.d. × 300 mm. The total surface area of the bellows is 850 cm$^2$ and the average thickness of the bellows is about 0·15 mm. The interior of the separator is evacuated and outside the bellows, sodium flows at a maximum temperature of 700°C.

By using such a bellows, in addition to increasing the mechanical strength, the sensitivity for the hydrogen detection is increased immensely. Furthermore, because of the large surface area, it is thought possible to use it as a hydrogen separator.

In the preliminary stage, the following interesting results have been obtained for type 316L bellows:

(i) The hydrogen introduced into sodium through this bellows is easily cold trapped. With our cold trap (cf. main paper) sodium hydride, as well as sodium oxide, is trapped with 100% yield.
(ii) The partial pressure of hydrogen in equilibrium with sodium, for the mother loop plugging temperature of $116 \pm 5$°C and the hydrogen separator temperature of 700°C, is about 0·04 mm Hg.
(iii) For the above condition, the hydrogen removal rate, with about $10^{-3}$ mm Hg vacuum, is $1{\cdot}5 \times 10^{-2}$ $cm^2$ (s.t.p.)/h.
(iv) Analysis of the exhaust gas was carried out by gas chromatography.

The total surface area can easily be increased one hundredfold by installing many bellows in parallel inside a cylinder about 250 × 1000 mm. Furthermore, by using Ni, the removal rate can be increased fivefold,[S3] resulting in a design with a removal rate of 0·02 g hydrogen/day or 0·5 g NaH/day. As the hydrogen saturation solubility at 110°C is in the order of magnitude of $10^{-3}$ ppm[S4] the above is a practical method of removal.

For the rapid removal of hydrogen from highly contaminated sodium, cold trapping is superior, but by installing this device as well, the following advantages can be obtained:

(i) Higher hydrogen (or water-leak) detection sensitivity.
(ii) Better hydrogen removal capability than a cold trap.
(iii) Wear free and plug free, permanent operation.
(iv) NaH can be removed from the cold trap, increasing the life of the cold trap.

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